Experimental Techniques: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Differential magnitude. Return →

$\Delta m = -2.5\log_{10}(100/200) = -2.5\log_{10}(0.5) = 0.753$ mag. The target is fainter than the comparison. **Check:** a factor of two in flux is about 0.75 mag.

Solution 2 — Eclipse depth. Return →

The flux ratio is 0.9, so the depth is $-2.5\log_{10}(0.9) = 0.114$ mag. **Check:** a 10% flux drop is slightly more than 0.1 mag.

Solution 3 — Comparison-star choice. Return →

A good comparison is stable, similar in brightness and colour to the target, close enough to share the same airmass and transparency, and free from crowding or detector defects. **Check:** multiple comparisons allow a consistency test.

Solution 4 — Common trend. Return →

The differential magnitude is unchanged because both fluxes are multiplied by the same factor, and the ratio cancels. **Check:** this is the main advantage of differential photometry.

Solution 5 — Phase. Return →

$6.25/2.5 = 2.5$ cycles, so the phase is 0.5. **Check:** phase is the fractional part of the cycle count.

Solution 6 — Calibration order. Return →

Bias removes the electronic zero level, dark removes thermal signal, and flat removes pixel-to-pixel sensitivity and illumination patterns. They are applied before analysis so instrumental structure is not mistaken for astrophysical signal. **Check:** calibration is part of the measurement, not an optional decoration.

Solution 7 — Significance. Return →

The depth is 3 times the scatter, which is suggestive but not automatically convincing. Check the comparison stars, airmass trend, flat-field stability, focus, and whether the dip repeats at the predicted phase. **Check:** a real eclipse should not appear in stable comparison stars.

Solution 8 — Cadence. Return →

The cadence is $90/15 = 6$ minutes per point. Observing outside eclipse establishes the baseline, reveals trends and allows the depth to be measured relative to a stable out-of-eclipse level. **Check:** ingress and egress may require even finer cadence.

Solution 9 — False signal. Return →

A simultaneous dip in target and comparison is likely instrumental or atmospheric: a cloud, tracking glitch, focus change, flat-field error or guiding issue. Test with additional comparison stars, check weather and guiding logs, and repeat on another night. **Check:** a true target eclipse should not appear in unrelated stable stars.

Solution 10 — Design an experiment. Return →

Choose two or three stable comparison stars of similar magnitude and colour within the field. Take bias, dark and flat frames, and use a fixed filter and exposure time that keeps the target well below saturation while giving high SNR. Observe before, during and after the predicted eclipse with a cadence of a few

minutes. Use a fixed photometric aperture or consistent PSF fitting, monitor airmass, and check that the comparison stars remain constant. Control flat-field and colour-dependent extinction systematics. Report the differential light curve, residual scatter, depth and uncertainty. **Check:** the experiment is designed to distinguish a real eclipse from common-mode noise.